LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Practice Pack: Sound

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Level 1 checks the ideas, Level 2 is board-style numericals, and Level 3 is Foundation-level challenge.
2. Key relations: v = fλ (speed = frequency × wavelength) · f = 1/T · echo distance = (speed × time)/2

LEVEL 1: CONCEPTS

Q1. Why can sound not travel through a vacuum?

Q2. Define frequency and give its SI unit.

Q3. What is the audible range of frequencies for human beings?

Q4. Which property of sound decides its (a) loudness and (b) pitch?

Q5. Why are the ceilings of concert halls and auditoriums curved?

LEVEL 2: NUMERICALS

Q6. A sound wave has a frequency of 2 kHz and a wavelength of 35 cm. How long will it take to travel 1.5 km?

Q7. A person claps near a cliff and hears the echo after 2 s. How far is the cliff? (Speed of sound = 346 m/s)

Q8. A source vibrates at 100 Hz. How many times does it vibrate in one minute?

Q9. A sonar on a submarine receives the echo of its signal 5 s after sending it. The object is 3625 m away. Find the speed of sound in water.

Q10. Find the wavelength of a 220 Hz sound in air, where its speed is 440 m/s.

LEVEL 3: CHALLENGE (FOUNDATION)

Q11. A stone is dropped from the top of a 500 m tall tower into a pond at its base. After how long is the splash heard at the top? (g = 10 m/s^2, speed of sound = 340 m/s)

Q12. What is the minimum distance needed between you and a wall to hear a distinct echo? (The sensation of a sound lasts about 0.1 s; speed of sound = 344 m/s.)

Q13. Bats fly in the dark without hitting objects. Explain how they do it.

~ Echo questions: remember to halve the total distance! ~